An automorphism inverting most elements
Herstein, 2.5.52

Theorem 1. *Let G be a finite group and let φ be an automorphism of G such that $\varphi(x) = x^{-1}$ for more than three-fourths of the elements of G. Then $\varphi(y) = y^{-1}$ for all $y \in G$, and hence G is abelian.*

Proof. Let

$$T = \{x \in G \mid \varphi(x) = x^{-1}\}.$$

The hypothesis is that

$$3|G| < 4|T|.$$

We first note a counting fact. If $A, B \subseteq G$ satisfy $|A| = |B| = c$ and $|G| < 2c$, then $A \cap B$ is nonempty. Indeed,

$$|A \cup B| + |A \cap B| = |A| + |B| = 2c,$$

whereas $|A \cup B| \leq |G| < 2c$, so $|A \cap B| > 0$.

We show that every element of T is central. Fix $x \in T$ and $g \in G$. Thus $\varphi(x) = x^{-1}$. Let

$$A = xT = \{xy \mid y \in T\}.$$

Left multiplication by x is injective, so $|A| = |T|$. Since

$$|A \cup T| + |A \cap T| = |A| + |T| = 2|T|$$

and $|A \cup T| \leq |G|$, we obtain

$$|A \cap T| \geq 2|T| - |G|.$$

Consequently, using $3|G| < 4|T|$,

$$2|A \cap T| \geq 4|T| - 2|G| > |G|,$$

that is,

$$|G| < 2|A \cap T|.$$

Every element $z \in A \cap T$ commutes with x. Indeed, write $z = xy$ with $y \in T$. Since also $z \in T$, we have $\varphi(y) = y^{-1}$ and $\varphi(z) = z^{-1}$. Hence

$$x^{-1}y^{-1} = \varphi(x)\varphi(y) = \varphi(z) = z^{-1} = (xy)^{-1} = y^{-1}x^{-1}.$$

Thus $x^{-1}y^{-1} = y^{-1}x^{-1}$. Taking inverses, and using $(yx)^{-1} = x^{-1}y^{-1}$ and $(xy)^{-1} = y^{-1}x^{-1}$, gives $yx = xy$. Therefore

$$zx = (xy)x = x(yx) = x(xy) = xz.$$

Now consider $A \cap T$ and $g(A \cap T) = \{gu \mid u \in A \cap T\}$. Left multiplication by g is injective, so both sets have cardinality $|A \cap T|$. By the counting fact above and the inequality $|G| < 2|A \cap T|$, their intersection is nonempty. Choose

$$u_1 \in (A \cap T) \cap g(A \cap T).$$

Then there is $u_2 \in A \cap T$ such that $u_1 = gu_2$. It follows that

$$g = u_1 u_2^{-1}.$$

Both u_1 and u_2 lie in $A \cap T$, so they commute with x. Also, if some z commutes with x, then z^{-1} commutes with x: indeed,

$$z(z^{-1}x) = x = z(xz^{-1}),$$

where the second equality uses $zx = xz$, and cancelling z gives $z^{-1}x = xz^{-1}$. Applying this to u_2, we compute

$$\begin{aligned} gx &= u_1 u_2^{-1} x \\ &= u_1 (u_2^{-1} x) \\ &= u_1 (x u_2^{-1}) \\ &= (u_1 x) u_2^{-1} \\ &= (x u_1) u_2^{-1} \\ &= x (u_1 u_2^{-1}) \\ &= xg. \end{aligned}$$

Thus $xg = gx$. Since $g \in G$ was arbitrary, every element of T is central.

Next, every element of G is a product of two elements of T. Let $g \in G$. From $3|G| < 4|T|$ we have $|G| < 2|T|$. The set

$$\{gy^{-1} \mid y \in T\}$$

has cardinality $|T|$, since the map $y \mapsto gy^{-1}$ is injective. By the counting fact, it meets T nontrivially. Hence there exist $x, y \in T$ such that $x = gy^{-1}$. Then

$$xy = gy^{-1}y = g,$$

so $g = xy$ with $x, y \in T$.

Finally, let $g \in G$ and write $g = xy$ with $x, y \in T$. Since $x \in T$, we have already shown that x is central, so in particular $xy = yx$. Therefore

$$\begin{aligned} \varphi(g) &= \varphi(x)\varphi(y) \\ &= x^{-1} y^{-1} \\ &= (yx)^{-1} \\ &= (xy)^{-1} \\ &= g^{-1}. \end{aligned}$$

Thus $\varphi(g) = g^{-1}$ for all $g \in G$. In particular, every element of G now lies in T, and every element of T is central. Hence for all $a, b \in G$, we have $ab = ba$, so G is abelian. $\square$